SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT ENGINEERING

Academic Year: 2025-26

NARSEE MONJEE INSTITUTE OF MANAGEMENT STUDIES
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Program: B Tech/ MBA Tech/ B Tech Intg (CSE-CS/AI&ML/AI&DS/IT/CIVIL/CS/MECH/COMP/DS/AI/CSBS/CSE-DS(7057)/EXTC

Subject: Management Accounting for Engineers Year: I/II/IV Semester I/II/IV/VIII

Date : 02 July 2026 Time: 2.00 pm to 5.00 pm (3 hrs)

Marks: 100 No. of Pages: 5

Re-Exam (2022-23/2023-24/2024-25/2025-26)

Instructions:

1. This question paper contains 5 pages.
2. This question paper contains 5 pages
3. Answer to each new question to be started on a fresh page
4. Figure on the right hand side indicates full marks
5. Question no.1 is compulsory
6. Out of remaining questions, attempt any 4 questions
7. In all 5 questions to be attempted
8. Assume suitable data if necessary
9. Scientific calculator/simple calculators are allowed

CO / BL	Part	Question	Marks
Q1			**[20]**
CO-1 *BL-Remember*	a	Explain the different Branches of Accounting (Financial Accounting, Cost Accounting, and Management Accounting) and state the significance of each branch in the context of business decision-making.	(5)
CO-5 *BL-Apply*	b	From the following particulars, calculate: (a) Break-even point (in ₹) (b) Margin of safety (c) Sales required to earn a profit of ₹ 2,40,000 (d) Profit when sales are ₹ 15,00,000 (e) Margin of safety if the company earns a profit of ₹ 3,00,000 **Given: Fixed Cost = ₹ 2,40,000 | Profit = ₹ 60,000 | Sales = ₹ 6,00,000**	(5)
CO-1 *BL-Remember*	c	Define Management Accounting and explain its objectives and scope. How is it different from Cost Accounting?	(5)
CO-2 *BL-Apply*	d	A manufacturing company faces inventory management issues. The store manager provides the following data for a critical raw material:	(5)

Particulars	Data
Normal usage	400 units per week
Maximum usage	600 units per week
Minimum usage	250 units per week
Lead time	3 to 5 weeks (normal: 4 weeks)
Re-order quantity	2,000 units

		Required: Calculate (i) Re-order Level (ii) Minimum Level (iii) Maximum Level (iv) Average Stock Level. (v) Also calculate the Economic Order Quantity (EOQ) if the ordering cost is ₹ 125 per order and the annual carrying cost is 20% of the purchase price of ₹ 25 per unit.	
Q2			**[20]**
CO-2 *BL-Remember*	**a**	Classify costs on the basis of their ELEMENTS and explain each element with suitable examples from a manufacturing company.	(5)
CO-2 *BL-Apply*	**b**	From the following particulars, prepare a cost statement showing all components of total cost and profit for the year ended 31st March 2026. **Units Produced = 8,000 | Units Sold = 7,000 | Selling Price = ₹ 120 per unit**	(15)

Particulars	1-4-2025 (₹)	Particulars	31-3-2026 (₹)
Stock of finished goods	Nil	Stock of finished goods (closing)	?
Stock of raw materials	45,000	Stock of raw materials	60,000
Work-in-progress	30,000	Work-in-progress	30,000
Purchases of raw materials	3,20,000	Director's Fees	12,000
Carriage inwards	15,000	Sales for the year	8,40,000
Productive Wages	1,60,000	Income Tax Paid	35,000
Factory Manager's Salary	16,000	Dividend Paid	20,000
Indirect Wages	24,000	Loss on Sale of Plant	12,000
Factory Power	16,000	Depreciation on Office Furniture	4,000
Factory Rent and Insurance	32,000		
Depreciation on Plant & Machinery	24,000		
Drawing Office Salaries	8,000		
Royalty on Production	40,000		
Office Rent and Insurance	16,000		
Audit Fees	8,000		
Salesman's Salary and Commission	28,000		
Carriage Outwards	14,000		
Advertisement	21,000		
Depreciation on Delivery Van	7,000		

<table>
<tr><td></td><td></td><td>Closing stock of finished goods to be valued at Cost of Production. Prepare the cost sheet showing total cost and cost per unit at each stage and also find total profit and profit per unit. (Note: Exclude items not related to cost of production.)</td><td></td></tr>
<tr><td colspan="3">Q3</td><td>[20]</td></tr>
<tr><td>CO-5
BL-Apply</td><td>a</td><td>The following is the budget for the year 2026:
<table>
<tr><th>Particulars</th><th>Amount (₹)</th></tr>
<tr><td>Sales (80,000 units @ ₹ 25)</td><td>20,00,000</td></tr>
<tr><td>Variable Cost</td><td>12,00,000</td></tr>
<tr><td>Contribution</td><td>8,00,000</td></tr>
<tr><td>Fixed Cost</td><td>3,00,000</td></tr>
<tr><td>Net Profit</td><td>5,00,000</td></tr>
</table>
From the above set of information find out:
(a) The adjusted profits for 2026 under the following two plans and suggest which plan should be implemented:
<table>
<tr><th>Change</th><th>Plan A</th><th>Plan B</th></tr>
<tr><td>Selling price</td><td>Increase by 15%</td><td>Decrease by 10%</td></tr>
<tr><td>Volume</td><td>Decrease by 20%</td><td>Increase by 25%</td></tr>
<tr><td>Variable cost per unit</td><td>Increase by 10%</td><td>Decrease by 5%</td></tr>
<tr><td>Fixed cost</td><td>Increase by 5%</td><td>No change</td></tr>
</table>
(b) The P/V ratio and break-even points under the two plans referred above.</td><td>(15)</td></tr>
<tr><td>CO-3
BL-Understand</td><td>b</td><td>Explain the different types of budgets prepared by an organisation. Briefly describe any FIVE types with suitable examples.</td><td>(5)</td></tr>
<tr><td colspan="3">Q4</td><td>[20]</td></tr>
<tr><td>CO-2
BL-Apply</td><td>a</td><td>A company currently purchases a raw material in lots of 500 units, which represents a 3-month supply. The cost per unit is ₹ 40, the ordering cost is ₹ 100 per order, and the inventory carrying cost is 25% per year of the unit cost.
You are required to:
1. Calculate the Economic Order Quantity (EOQ).
2. Calculate the annual savings in total inventory cost if the company shifts to ordering at EOQ instead of the present order quantity.
3. Calculate the number of orders per year and time gap between consecutive orders (in days) at EOQ level.</td><td>(12)</td></tr>
<tr><td>CO-5
BL-Apply</td><td>b</td><td>Explain Cost-Volume-Profit (C-V-P) Analysis. State the key assumptions underlying CVP analysis and explain how it is used by management for short-term decision-making.</td><td>(8)</td></tr>
<tr><td colspan="3">Q5</td><td>[20]</td></tr>
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<table>
<tr><td>CO-3
BL-Remember</td><td>a</td><td>Sunlight Manufacturing Ltd. is currently operating at 60% capacity (12,000 units). You are required to prepare a Flexible Budget for 80% and 100% capacity. Consider the following information:
<table><tr><th>Particulars</th><th>Nature</th><th>Amount at 60% (₹)</th></tr><tr><td>Sales</td><td>–</td><td>24,00,000</td></tr><tr><td>Material</td><td>Variable</td><td>6,00,000</td></tr><tr><td>Labour</td><td>Variable</td><td>3,60,000</td></tr><tr><td>Variable Overheads</td><td>Variable</td><td>2,40,000</td></tr><tr><td>Fixed Overheads</td><td>Fixed</td><td>4,80,000</td></tr><tr><td>Variable Selling Expenses</td><td>Variable</td><td>1,20,000</td></tr><tr><td>Fixed Selling Expenses</td><td>Fixed</td><td>1,80,000</td></tr><tr><td>Distribution Overheads (60% Fixed)</td><td>Semi-variable</td><td>1,20,000</td></tr></table>Adjustments:
i. Raw material cost per unit will increase by 8% when production reaches 80% capacity and above.
ii. Direct labour cost per unit will increase by 10% when capacity exceeds 75%.
iii. Variable factory overheads will decrease by 5% per unit when production exceeds 85% capacity (efficiency gains).
iv. Selling price = ₹ 200 per unit; a discount of 3% will be allowed at 80% capacity and above.
v. Variable selling expenses will increase by ₹ 3 per unit when capacity reaches 80% and above.
vi. An additional fixed overhead of ₹ 60,000 will be incurred at 100% capacity.</td><td>(15)</td></tr>
<tr><td>CO-3
BL-Apply</td><td>b</td><td>What is a Flexible Budget? Explain its advantages over a Fixed Budget with suitable examples.</td><td>(5)</td></tr>
<tr><td colspan="3">Q6</td><td>[20]</td></tr>
<tr><td>CO-4
BL-Apply</td><td>a</td><td>The standard cost of a product shows the following:
Material : 3 kg @ ₹ 15 per kg
Wages : 4 hours @ ₹ 30 per hour

The actual data from business operations:
Production : 8,000 units
Material consumed : 25,000 kg for ₹ 3,62,500
Wages paid : 30,500 hours for ₹ 9,45,500

Calculate:
a) Material Cost Variance, Material Usage Variance and Material Price Variance
b) Labour Cost Variance, Labour Rate Variance and Labour Efficiency Variance</td><td>(12)</td></tr>
<tr><td>CO-3
BL-Understand</td><td>b</td><td>What is Standard Costing? Explain the steps involved in establishing a Standard Costing system. Also state the advantages of Standard Costing to management.</td><td>(8)</td></tr>
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<table>
<tr><td colspan="3">Q7</td><td>[20]</td></tr>
<tr><td>CO-1
BL-Remember</td><td>a</td><td>For the production of 12,000 electric fans, the following are the budgeted expenses:
<table><tr><th>Particulars</th><th>₹ Per Unit</th></tr><tr><td>Direct Material</td><td>50</td></tr><tr><td>Direct Labour</td><td>24</td></tr><tr><td>Variable Overheads</td><td>16</td></tr><tr><td>Fixed Overheads (₹ 2,88,000 total)</td><td>24</td></tr><tr><td>Variable Expenses (Direct)</td><td>10</td></tr><tr><td>Selling Expenses (25% Fixed)</td><td>20</td></tr><tr><td>Administrative Expenses (₹ 72,000 Fixed)</td><td>6</td></tr><tr><td>Distribution Expenses (40% Fixed)</td><td>10</td></tr><tr><td>Total Cost per Unit</td><td>160</td></tr></table>
Prepare a budget for the production of 8,000, 10,000 and 15,000 fans, showing total cost and cost per unit at each level.</td><td>(15)</td></tr>
<tr><td>CO-3
BL-Understand</td><td>b</td><td>Discuss the advantages of budgeting for a business organisation. How does budgeting help management in planning, coordination, and control of business activities?</td><td>(5)</td></tr>
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